From Bars to Analytic HFT Support

The trajectory–kernel route to the TK6B.5f all-orders theorem

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Abstract

Bars proposed that at strong coupling the perturbative massive primaries of the type-IIA superstring should be supplemented by nonperturbative states so that their cumulative Spin(9) content assembles into Spin(10) representations. In its minimal form the claim is an existence problem: for every level n, the cumulative character $R_n^9 = \sum_{i=1}^n r_i^9$ should admit at least one positive integral Spin(10) lift. The lift need not be unique.

This report first develops the full route from Bars's proposal to the local all-orders theorem proved in TK6B.5f. It reviews the covariant HFT generating function, multiplicity-free $D_5 \downarrow B_4$ branching, Regge shifts, restriction-kernel coordinates, positive packet cones, Gale circuits, interlacing shells, the local Möbius inverse and stable Pieri translation. The mathematical notions beyond a standard graduate representation-theory course are introduced when they enter.

The second part proves the HFT support theorem. A finite triality identity telescopes the Green–Schwarz oscillator product and performs the universal-supermultiplet division before support estimates are applied. A Weyl-numerator argument transfers Spin(8) support to Spin(9), uniform vector and spinor estimates close the infinite oscillator tail, and a fixed 323-record low-mode certificate finishes the proof. This closes the HFT premise needed for stable Pieri transport at every degree. The remaining Bars problem is global: control all moving weighted circuits and prove integral packetization at arbitrary level.

Three levels of the argument.
Bars's target: for every n, find a positive integral Spin(10) character X_n with

$$X_n{\downarrow_{\mathrm{Spin}(9)}} = R_n^9 = \sum_{i=1}^{n} r_i^9.$$

The theorem proved here: if the irreducible Spin(9) character V_μ occurs with nonzero signed coefficient in degree d of the virtual HFT remainder $F(q)$, then

$$[V_\mu]F_d \neq 0 \quad \Longrightarrow \quad d \geq K(\mu),$$

where K is the exact Pieri-hazard threshold derived in TK6B.5e. The theorem holds at every oscillator degree.
The remaining gap: show that stable transport controls every horizon-dependent weighted circuit and supply an all-level integral packet decomposition (or a saturation theorem). Finite levels are evidence and regression data, not the all-orders argument.

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1 Bars's strong-coupling proposal

1.1 Why Spin(9) and Spin(10) appear

For a massive particle in ten-dimensional Minkowski space the little group is Spin(9). Perturbative massive string states therefore decompose into Spin(9) irreducible representations. If strongly coupled type-IIA string theory develops an eleventh spacetime dimension, the corresponding massive rotation group is Spin(10). A necessary representation-theoretic shadow of that claim is that the ten-dimensional states, after adding appropriate nonperturbative states, should be restrictions of Spin(10) representations.

The supersymmetry descendants are conveniently factored off. Let r_n^9 denote the Spin(9) character of the chiral massive *primaries* at perturbative level n. Acting with the broken supercharges produces the universal chiral massive multiplet

$$Z_Q = [2,0,0,0]_9 + [1,0,0,1]_9 + [0,0,1,0]_9,$$

of dimension $44 + 128 + 84 = 256$. For the closed type-II theory there is one such factor from each chiral half. Bars's question concerns the remaining primary factor: can it also be organized covariantly with respect to the extra spatial direction? This is the content relevant to the present project, rather than a claim that the perturbative spectrum alone already has Spin(10) symmetry.

We use Spin rather than SO when spinor representations matter. The older literature often writes $SO(9)$ and $SO(10)$ for the same character problem.

1.2 Bars's additional level and the cumulative sum

Bars introduces an elementary string level l and an additional nonperturbative level l'. At fixed ten-dimensional mass level n, his on-shell proposal is

$$n = l' + l,$$

with the pairs

$$(l', l) = (0, n), (1, n-1), \ldots, (n-1, 1).$$

The Spin(9) quantum numbers depend on l, while l' distinguishes the new states. In the single-multiplicity, or "minimal", scheme each pair is used once. Consequently one chiral half at mass level n has the cumulative primary character

$$R_n^9 = \sum_{l=1}^{n} r_l^9. \tag{1}$$

Bars checked directly through $n = 5$ that these cumulative characters reassemble into Spin(10) characters [1]. For orientation, his dimension notation gives the following first examples. Bosonic and fermionic sectors are displayed separately where needed.

Table 1: Bars's minimal cumulative pattern for one chiral half. These are the examples displayed in the 1995 paper, not an all-orders proof.

n	cumulative Spin(9) primary content	one Spin(10) parent
1	1	1
2	$1+9$	10
3	$(1+9+44)_B+16_F$	54_B+16_F
4	$(1+2\cdot 9+36+156+44)_B+(16+128)_F$	$(45+210)_B+144_F$
5	$\big(2(1+9+36+44)+84+156+231+450\big)_B$ $+\big(2(16+128)+576\big)_F$	$(1+120+320+660)_B+(144+720)_F$

The representation dimensions in the last column suppress Dynkin labels and, for spinors, an outer-chirality choice. Later finite computations show that positive parents need not be unique. Bars's Section III.B claim therefore has the following precise minimal interpretation.

Definition 1.1 (Minimal Bars lifting problem)**.** For every $n \geq 1$, find at least one positive integral Spin(10) character X_n such that

$$\mathrm{Res}^{\mathrm{Spin}(10)}_{\mathrm{Spin}(9)}\, X_n = R^9_n.$$

"Positive integral" means a finite direct sum of irreducible characters with coefficients in $\mathbb{Z}_{\geq 0}$. It does not mean that the lift is unique, nor that its consecutive difference $X_n - X_{n-1}$ is positive. It also does not include Bars's separate higher-multiplicity scheme or arbitrary additional complete nonperturbative multiplets. The protected project data through level 10 provide examples of such lifts; only levels 1–5 were displayed by Bars himself.

2 The HFT covariant spectrum

2.1 Characters and plethystic exponentials

Hanany, Forcella and Troost (HFT) supplied the all-order perturbative input which Bars did not have: a generating function covariant under the massive little group Spin(9) [2]. We write

$$[a_1, a_2, a_3, a_4]_9$$

for the irreducible $B_4 = \mathrm{Spin}(9)$ character with those Dynkin labels and set

$$V_9 = [1,0,0,0]_9, \qquad S_9 = [0,0,0,1]_9.$$

On the common Spin(8) torus the physical transverse vector is represented by the virtual B_4 character

$$T = V_9 - \mathbf{1},$$

because V_9 restricts as $8_v \oplus \mathbf{1}$.

For a character-valued series f, the bosonic and fermionic plethystic exponentials are

$$\mathrm{PE}[f] = \exp\left(\sum_{k\geq 1} \frac{f(q^k, y^k)}{k}\right),$$

$$\mathrm{PE}_F[f] = \exp\left(\sum_{k\geq 1} \frac{(-1)^{k+1} f(q^k, y^k)}{k}\right).$$

They are compact devices for taking all symmetric, respectively exterior, powers of oscillator representations. The variables collectively denoted by y are weight fugacities.

2.2 The HFT formula

Define the transverse bosonic product

$$Z_B(q) = \mathrm{PE}\left[\frac{q}{1-q}T\right]$$

and the fermionic product with a fermion-mode fugacity f,

$$\mathcal{Z}_F(q; f) = \mathrm{PE}_F\left[\frac{f}{1-q}T\right].$$

The GSO-projected Neveu–Schwarz series is

$$Z_{NS}(q) = \frac{1}{2\sqrt{q}}\left(\mathcal{Z}_F(q; \sqrt{q}) - \mathcal{Z}_F(q; -\sqrt{q})\right),$$

while the Ramond oscillator factor entering the covariant expression is

$$Z_R(q) = \mathcal{Z}_F(q; q).$$

With a purely massless chirality term denoted by Δ_0, HFT's covariant left-moving partition function can be written

$$Z_{\mathrm{Left}}(q) = \Delta_0 + Z_B(q)\left(Z_{NS}(q) + \frac{1}{2}S_9 Z_R(q)\right). \tag{2}$$

The apparently fractional factor does not produce fractional multiplicities: the plethystic expansion supplies the compensating parity factor.

Let Z_0 be the complete massless contribution. The massive-primary series is HFT's quotient

$$Z_m(q) = \frac{Z_{\mathrm{Left}}(q) - Z_0}{qZ_Q}. \tag{3}$$

In project notation,

$$R(q) = Z_m(q) = \sum_{d\geq 0} r^9_{d+1} q^d. \tag{4}$$

Thus coefficient degree $d = 0$ corresponds to physical massive level one. Equation (3) is an exact all-orders character formula, but positivity of its irreducible Spin(9) coefficients is not manifest from the quotient. HFT themselves identified proving positivity as a challenge. For the Bars problem, $R(q)$ supplies the single-level characters which are then cumulatively summed as in (1).

2.3 Removing the leading symmetric-vector tower

HFT also observed the exact factorization

$$R(q) = \mathrm{PE}(qV_9)F(q), \qquad F(q) = \sum_{d\geq 0} F_d q^d. \tag{5}$$

The coefficient of the first factor is

$$[q^a]\,\mathrm{PE}(qV_9) = \mathrm{Sym}^a(V_9) = \bigoplus_{j=0}^{\lfloor a/2\rfloor} [a-2j,0,0,0]_9.$$

This factor carries the leading Regge tower. The coefficients F_d are *virtual* characters: irreducible coefficients may be negative because we have algebraically removed nine level-one modes even though only eight are physical transverse oscillators. Therefore an argument about F must respect cancellations; F_d is not the character of a Hilbert space.

Equation (5) is the entry point of the final proof, but several layers of the lifting problem had to be understood before the right support statement for F became visible.

3 Branching packets and positive lifts

3.1 The multiplicity-free branching rule

Write a dominant $D_5 = \mathrm{Spin}(10)$ highest weight in orthogonal coordinates as

$$\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq \lambda_4 \geq |\lambda_5|$$

and a dominant $B_4 = \mathrm{Spin}(9)$ weight as

$$\mu_1 \geq \mu_2 \geq \mu_3 \geq \mu_4 \geq 0.$$

The restriction $D_5 \downarrow B_4$ is multiplicity-free. The child V_μ occurs in V_λ precisely when the coordinates interlace:

$$\lambda_1 \geq \mu_1 \geq \lambda_2 \geq \mu_2 \geq \lambda_3 \geq \mu_3 \geq \lambda_4 \geq \mu_4 \geq |\lambda_5|. \tag{6}$$

Definition 3.1 (Branching packet)**.** The packet of a D_5 irrep V_λ is the B_4 character

$$P_\lambda = \mathrm{Res}^{D_5}_{B_4} V_\lambda,$$

namely the sum of all interlacing children in (6). Its multiplicity vector has only zeroes and ones.

A positive integral lift of a target multiplicity vector m is exactly a nonnegative integral packet decomposition

$$m = Ax, \qquad x \in \mathbb{Z}^p_{\geq 0},$$

where the columns of A are the permitted packets. This converts Bars's question into a structured integer decomposition problem.

For a top B_4 Dynkin label $\mu = [\ell, m, r, s]_9$, the unoriented family of parents with that same top child is

$$[\ell, m, r, k, s-k]_{10}, \qquad 0 \leq k \leq \left\lfloor \frac{s}{2} \right\rfloor. \tag{7}$$

Outer-conjugate D_5 parents restrict identically, so they may be collapsed when the question is existence rather than chirality selection.

3.2 A triangular virtual section

Choose the $k = 0$ parent

$$p_0(\ell, m, r, s) = [\ell, m, r, 0, s]_{10}$$

for every B_4 top weight. Ordered by dominance, the resulting packet matrix U is triangular with diagonal entries one. Hence

$$\det U = 1$$

in every finite downward-closed domain, and any integral target has a unique integral *virtual* lift on this section:

$$c = U^{-1}m.$$

Some coefficients c_μ can be negative. The role of alternative parents is precisely to repair those negative coefficients while preserving the restriction.

Definition 3.2 (Restriction kernel and positive fibre)**.** The restriction kernel consists of virtual D_5 characters K satisfying $\operatorname{Res} K = 0$. If $s(m)$ is any virtual section lift, every other lift is

$$s(m) + K.$$

The *positive fibre* over m is the set of those kernel corrections for which every final D_5 coefficient is a nonnegative integer.

This language separates existence from selection. Bars III.B asks whether the positive fibre is nonempty. It does not ask for a preferred point in that fibre.

4 The trajectory–kernel programme

4.1 Step 1a: Regge coordinates and shifts

At each physical HFT level $n \geq 1$, write

$$a_n(\ell, m, r, s) = \operatorname{mult}_{r_n^9}[\ell, m, r, s]_9.$$

Thus a five-tuple (n, ℓ, m, r, s) means "level n, with B_4 highest weight $[\ell, m, r, s]_9$". The first Dynkin coordinate ℓ measures the leading symmetric-tensor direction. The basic Regge shift is

$$(n, \ell, m, r, s) \longmapsto (n + 1, \ell + 1, m, r, s). \tag{8}$$

A *Regge column* fixes (m, r, s). The elementary orbit invariant is therefore

$$\rho = n - \ell.$$

At fixed (m, r, s), a Regge orbit is therefore specified by ρ. Nothing about the shift itself singles out the further subtraction of $3m$; that choice becomes natural only after the support geometry is introduced.

4.2 Step 1b: HFT support and the GMSZ chambers

Through the tested finite window, the HFT data are accurately organized by the empirical support rule of Georgoudis, Minahan, Ström and Zoumis (GMSZ) [3]. Their rule is stated for $n > 6$. Define the support cost

$$c(r, s) = \max\{6r + 5s - 6,\ 6r + 4s - 2,\ 6r + 3s - 1,\ 2s + 1\}.$$

At fixed (n, m, r, s), the largest allowed leading label is

$$\ell_{\max} = n - 3m - c(r, s),$$

apart from the stated isolated gap $[n-2, 0, 0, 0]_9$. Thus the outer support condition is

$$0 \le \ell \le \ell_{\max}.$$

This is what motivates the support-adapted reduced depth

$$\delta = n - \ell - 3m = \rho - 3m.$$

The subtraction of $3m$ is not required by Regge invariance. It absorbs the three units of level carried by one unit of the second Dynkin label in the GMSZ support rule. The support condition becomes simply

$$\delta \ge c(r, s),$$

and the literal distance below the leading edge is

$$\ell_{\max} - \ell = \delta - c(r, s).$$

Thus $\rho = n - \ell$ is the basic Regge invariant, whereas δ is the same invariant recentered so that the m-dependent support cost has already been removed.

Some chamber terminology will be used repeatedly. A supported target with $\ell \ge 1$ has the unit Regge predecessor

$$(n-1, \ell-1, m, r, s).$$

In the GMSZ range, its *upper ℓ-chamber* is the high-ℓ part of the support for which that predecessor is present and

$$\ell > \left\lfloor \frac{\ell_{\max}}{2} \right\rfloor.$$

Here "upper" means closer to the largest allowed first Dynkin label; it does not mean higher string level. The multiplicity in this chamber is transported unchanged from the predecessor. The lattice wall at the edge of this region is the *transport boundary*. The *outer support boundary* is instead the edge of the allowed spectrum itself, such as $\ell = \ell_{\max}$ at fixed (n, m, r, s). A third object, the branching boundary of an interlacing packet, will appear in Step 2. These boundaries are related manifestations of failed naive translation, but they are not the same mathematical locus.

We can now state the three transport rules precisely in our conventions. Fix a supported target $[\ell, m, r, s]_9$ at level $n > 6$, and let $\ell_{\max}$ always mean the target value defined above. A proposed equality is admissible only when its lower-level source exists in the HFT support and the indicated positive integer k is within range.

The *ℓ-transport* is

$$a_n(\ell, m, r, s) = a_{n-k}(\ell - k, m, r, s), \qquad 1 \le k \le \ell,$$

with chamber inequality

$$\ell > \left\lfloor \frac{\ell_{\max}}{2} \right\rfloor + \left\lfloor \frac{k}{2} \right\rfloor.$$

For $k = 1$ this reduces to the upper-ℓ-chamber condition already given.

The *m-transport* is

$$a_n(\ell, m, r, s) = a_{n-3k}(\ell, m - k, r, s), \qquad 1 \le k \le m,$$

with chamber inequality

$$\ell > \ell_{\max} - m + k - 1.$$

The factor three in the level shift is the same factor that motivated the coordinate $\delta = n - \ell - 3m$.

The *r-transport* is

$$a_n(\ell, m, r, s) = a_{n-6k}(\ell, m, r-k, s), \qquad 1 \le k \le r,$$

with chamber inequality

$$\ell > \ell_{\max} - r + k - 1 + \epsilon(s),$$

where the spinor-label correction is

$$\epsilon(s) = \begin{cases} 2, & s = 0, \\ 1, & 1 \le s \le 3, \\ 0, & s \ge 4. \end{cases}$$

All three inequalities are strict inequalities between integers. In the deterministic TK5 chamber decomposition, the families are tried in the order ℓ, m, r, using the largest admissible k in the first applicable family. The finite audit also tested every admissible k, rather than only that selected representative.

The *residual core* is the set of supported labels for which none of these admissible transports applies. It is therefore the part of the spectrum whose multiplicity is not supplied by the transport identities. In particular, it is not merely a synonym for the complement of the upper ℓ-chamber: some labels below that chamber can still be transported in the m or r direction.

4.3 Step 1c: what TK1 and TK5 established

TK1 asked, before imposing any chamber condition, whether HFT multiplicities were simply constant along all Regge orbits.

They are not globally constant. Through level 10, 80 of 134 adjacent tests matched and 54 failed. TK5 later showed that the tested upper ℓ-chamber contains no such failures. In the later level-7–11 audit, all 95 admissible adjacent comparisons inside that chamber matched. The failures lie outside it, where crossing the transport boundary and encountering genuinely nontransported core data cannot be ignored.

This chamber decomposition explained the TK1 mismatches, but its status was still empirical and finite-level. It prevented a false all-orders conjecture based only on leading trajectories. The later Pieri argument derives directly from the exact HFT factorization the portion of stable transport actually needed for the all-orders support theorem.

4.4 Step 2: boundary-corrected kernel shifts

Let σ_D and σ_B increase the first D_5 and B_4 Dynkin labels respectively. Interlacing gives the exact identity

$$\mathrm{Res}(\sigma_D V_\lambda) = \sigma_B \,\mathrm{Res}(V_\lambda) + \partial V_\lambda, \tag{9}$$

where ∂V_λ is the nonnegative, multiplicity-free slice

$$\mu_1 = \lambda_2.$$

This slice is the *branching boundary*: it is the face of the $D_5 \downarrow B_4$ interlacing packet left behind by the shift. It is not the transport boundary just defined, although both phenomena express a failure of a naive shift to act everywhere without correction. The slice is independent of the shifted first label. Thus raw shifting does not commute with restriction, but its failure is an explicitly known lower face of the packet.

If s_{ch} denotes the triangular all-chiral section, define the corrected shift on a kernel element by

$$T(K) = \sigma_D K - s_{\mathrm{ch}}(\mathrm{Res}(\sigma_D K)). \tag{10}$$

Then $\mathrm{Res}\, T(K) = 0$. TK2 proved more: for every natural finite generator, $T(K)$ is exactly the next generator along its Regge trajectory. This is why the programme is called *trajectory–kernel*: the leading shift acts naturally only after the restriction-kernel boundary correction is included.

The first important seed is the six-term relation

$$[0,0,0,1,1]_{10} - [0,0,0,0,2]_{10} - [0,0,1,0,0]_{10} + [0,1,0,0,0]_{10} - [1,0,0,0,0]_{10} + [0,0,0,0,0]_{10},$$

whose restriction vanishes. In dimension notation it is

$$210 - 126 - 120 + 45 - 10 + 1.$$

This is the local algebra behind the first 126/210 parent ambiguity.

4.5 Step 3: positive fibres and the failure of uniqueness

TK3 used the natural kernel basis to parameterize the complete positive fibres in the level-10 HFT closure. The number of positive lifts at levels 1 through 10 was

$$1, 1, 1, 1, 1, 1, 2, 8, 72, 6057.$$

The first compulsory correction occurs at level 7 and uses the six-term seed above. The result establishes two logically different facts:

- the tested cumulative HFT targets do have positive integral lifts;
- positivity alone very quickly ceases to choose a unique lift.

TK4 and TK5 tested possible selection principles: minimum kernel population, minimum support, minimum corrected-shift defect and sources confined to an empirical residual chamber. These organize the finite solutions but do not produce a unique cutoff-stable rule. In particular, a lift optimized at one horizon may have to be abandoned to reach the optimum at the next horizon. This redirected the project from selecting a favourite lift to proving that the positive fibre is nonempty by intrinsic inequalities.

5 From a positive fibre to facet inequalities

5.1 The global ring kernel

Let 16_+ and 16_- be the two half-spinor characters of D_5 and put

$$J = 16_+ - 16_-.$$

The representation-ring restriction is surjective and has the exact principal kernel

$$\ker(R(\mathrm{Spin}(10)) \longrightarrow R(\mathrm{Spin}(9))) = (J). \tag{11}$$

Thus every virtual ambiguity is a multiple of one global chirality difference. This elegant ring statement does not solve positivity: multiplication by J introduces both signs. The positive problem lives in the representation *semiring*, where subtraction is unavailable.

5.2 Cone, semigroup, facets and Hilbert basis

For a finite packet matrix A, define

$$C = A\mathbb{R}^p_{\geq 0}, \qquad S = A\mathbb{Z}^p_{\geq 0}.$$

The cone C records rational positive packet combinations; the semigroup S records genuine integral combinations. An integral target can lie in C but outside S, producing an *integrality hole*.

A *facet* is a codimension-one face of C. Equivalently it is given by a primitive linear inequality

$$y^T m \geq 0$$

valid on every packet column. Membership in a full-dimensional rational cone is equivalent to satisfying all its facet inequalities. A *Hilbert basis* is the minimal finite set of lattice points which generates all lattice points in the cone by nonnegative integer sums. If the packet columns themselves form that Hilbert basis, then

$$S = C \cap \mathbb{Z}^d,$$

and the semigroup is called *saturated* or *normal*.

TK6B.1 found, in the complete level-11 closure,

$$A \in \mathbb{Z}^{73\times 84}, \qquad \operatorname{rank} A = 73,$$

with 2508 facets. The matrix is not totally unimodular, but its 84 packet columns are nevertheless exactly its Hilbert basis. Hence, in that finite domain, the facet inequalities are both necessary and sufficient for an integral positive lift. This was decisive conceptually, although it was not yet an all-orders saturation theorem.

5.3 Gale transforms and positive circuits

Directly listing thousands of facet inequalities obscures their origin. Let

$$K \in \mathbb{Z}^{p\times k}$$

have as columns an integral basis of $\ker A$, so $AK = 0$. The row vectors $g_j \in \mathbb{Z}^k$ of K are a *Gale transform* of the packet configuration. Informally, the packet columns describe what is generated; the Gale vectors describe all linear dependencies among those generators.

If y is a facet functional, its slack on the packet columns is

$$x = A^T y.$$

It satisfies

$$x \geq 0, \qquad K^T x = 0.$$

Conversely, because A has full row rank, every extreme ray of this nonnegative relation cone comes from a facet. A *positive Gale circuit* is a minimally supported relation

$$\sum_j x_j g_j = 0, \qquad x_j \geq 0,$$

with primitive integer coefficients. The word "circuit" is borrowed from matroid theory: it means that deleting any supported row destroys the dependence.

In triangular coordinates $c = U^{-1}m$, the corresponding liftability condition becomes the weighted Hall-type inequality

$$\sum_{j \in \mathrm{Ref}} x_j c_j \geq 0. \tag{12}$$

At level 11, 2508 facets compressed to only 40 circuit types. At level 13, new shell collisions produced genuine coefficient-two circuits. Therefore the correct all-orders object is a growing family of *weighted* circuits, not a fixed list of zero-one inequalities.

6 Shell identities and the bridge back to HFT

6.1 Universal local kernel relations

The finite kernel families eventually proliferate, but their local origin is uniform. For the parent family (7), let

$$P_k = [\ell, m, r, k, s-k]_{10}.$$

The difference of two consecutive parent packets is a single interlacing boundary slice. Inclusion–exclusion on that slice gives four all-chiral packets $Q_{1,k}, \ldots, Q_{4,k}$ and the universal six-term identity

$$\mathrm{Res}(P_k - P_{k-1} - Q_{1,k} + Q_{2,k} - Q_{3,k} + Q_{4,k}) = 0. \tag{13}$$

For example, the first split is

$$\begin{aligned} K^{(1)}_{\ell,m,r,s} = {} & [\ell, m, r, 1, s-1] - [\ell, m, r, 0, s] - [\ell, m, r+1, 0, s-2] \\ & + [\ell, m+r+1, 0, 0, s-2] + [m, r, 0, 0, s-2] - [\ell+m+1, r, 0, 0, s-2]. \end{aligned} \tag{14}$$

All labels in this display are D_5 labels. The identity follows directly from the interlacing rule and is valid for arbitrary nonnegative labels. Summing consecutive shells telescopes the kth parent back to the all-chiral parent with exactly $4k+2$ unit-coefficient terms.

The important point is where level dependence has gone. The shell formula is universal; a string-level cutoff merely chooses a finite induced collision hypergraph of these local relations. Its primitive positive circuits are the facet inequalities that the HFT target must satisfy.

6.2 The local Möbius inverse

The triangular section itself also has a cutoff-independent formula. Let

$$M(\ell, m, r, s)$$

be the multiplicity of $[\ell, m, r, s]_9$ in any finite B_4 character, extended by zero outside the dominant chamber. The coefficient of the all-chiral parent $[\ell, m, r, 0, s]_{10}$ is

$$\begin{aligned} c_{\ell,m,r,s} = {} & \sum_{\epsilon_1,\epsilon_2,\epsilon_3 \in \{0,1\}} (-1)^{\epsilon_1+\epsilon_2+\epsilon_3} \\ & M(\ell + \epsilon_1 - \epsilon_2, m + \epsilon_2 - \epsilon_3, r + \epsilon_3, s). \end{aligned} \tag{15}$$

This is a three-fold finite difference, or Möbius inversion, of the interlacing sums. "Möbius" here means inclusion–exclusion on a finite partially ordered set; no number-theoretic Möbius function is involved.

Substituting (15) into (12) turns every branching-cone facet into a finite signed inequality among HFT multiplicities. This is the essential bridge: an abstract Spin(10) lifting obstruction becomes an explicit local functional of the known Spin(9) generating series.

6.3 Why higher-level enumeration cannot finish the proof

The full infinite shell system has no nonzero finitely supported positive circuit. The reason is that one boundary term in (13) forgets the leading label ℓ, so a putative finite circuit can always be tested against a sufficiently distant shell and forced negative. Finite-horizon facets are therefore moving *boundary-tail* objects; their support changes as the horizon grows.

Some tails telescope under (15) to a bounded number of endpoint multiplicities, but mixed circuits can have growing pullback support. This is the precise answer to the question "what prevents an all-orders proof from being just a check of higher and higher levels?" One needs a structural transport theorem for the moving tails. That requirement led back to the HFT symmetric-vector factorization.

7 Stable Pieri translation and the remaining support theorem

7.1 What "stable Pieri translation" means

Let V_μ and V_λ be irreducible B_n representations, expressed in orthogonal coordinates, and let

$$H_k = V_{k\omega_1}$$

be the traceless symmetric, or harmonic one-row, representation. A Pieri rule describes the irreducible summands obtained by tensoring an arbitrary representation with such a one-row representation.

Theorem 7.1 (Stable Pieri translation)**.** *If the prospective target is at least as far out in its first orthogonal coordinate as the source,*

$$\lambda_1 \geq \mu_1,$$

then

$$[V_\lambda : V_\mu \otimes H_k] = [V_{\lambda+\omega_1} : V_\mu \otimes H_{k+1}] \tag{16}$$

for every $k \geq 0$. Consequently

$$[V_\lambda : V_\mu \otimes \mathrm{Sym}^a(V)] = [V_{\lambda+\omega_1} : V_\mu \otimes \mathrm{Sym}^{a+1}(V)]. \tag{17}$$

The brackets denote tensor-product multiplicity. In B_4, $\lambda \mapsto \lambda + \omega_1$ is exactly $[\ell, m, r, s] \mapsto [\ell + 1, m, r, s]$. Thus the theorem says that, once the first row is longer than anything carried by the fixed source V_μ, adding one symmetric-vector degree simply adds one box to that first row. The decomposition has entered a stable regime.

The proof uses Racah–Speiser. Every Weyl-group term which moves the first coordinate would require a weight of V_μ whose absolute coordinate is larger than μ_1, so it vanishes. The stabilizer of the first coordinate is unchanged when one adds ω_1 to the target and one unit to the harmonic degree. This works equally for integral tensor weights and half-integral spinor weights; stable Pieri translation itself is not a chirality obstruction.

7.2 Applying it to the HFT factorization

From (5), the coefficient of q^N is

$$[q^N]R(q) = \sum_{d=0}^{N} \mathrm{Sym}^{N-d}(V_9) \otimes F_d.$$

For a fixed V_μ in F_d, equation (17) transports its contribution from one physical level to the next provided the target satisfies $\lambda_1 \geq \mu_1$. The sign of the coefficient of V_μ in the virtual character F_d is irrelevant: the equality holds contribution by contribution before the signed terms are added.

Two possible failures remain:

1. a source V_μ might couple to an *unstable* target with $\lambda_1 < \mu_1$;
2. the new top-degree term F_N, which has no predecessor, might itself hit the translated target.

Both are excluded if every remainder label appears late enough.

7.3 The Pieri hazard

Define the exterior-chamber cost

$$E(\ell, m, r, s) = 2\ell + 3m + c(r, s),$$

where $c(r, s)$ is the support cost introduced above. For a source μ, define its harmonic hazard

$$H(\mu) = \max_{\substack{k\geq 0,\ V_\lambda \subset V_\mu \otimes H_k \\ \lambda_1 < \mu_1}} (E(\lambda) - k).$$

This maximum is finite. An unstable target can occur only for

$$k < 2\mu_1,$$

so no unbounded tensor-product search is hidden in the definition. Put

$$K(\mu) = \max(E(\mu) - 1, H(\mu)). \tag{18}$$

Then the single support implication

$$[V_\mu]F_d \neq 0 \quad \Longrightarrow \quad d \geq K(\mu) \tag{19}$$

simultaneously excludes all unstable couplings and the unpaired top-degree term.

The hazard is not an opaque maximization. Orthogonal Pieri interlacing [4] gives a closed piecewise-affine formula: on $\ell = 0$ it is simply $E - 1$, while for $\ell \geq 1$ three additional affine walls appear. The next section packages those pieces into a convenient seven-wall function K_7.

7.4 What the present proof accomplishes

Before TK6B.5f, implication (19) had been checked only through a finite HFT window. Checking more degrees could never establish it. The remaining task was to derive it from the infinite oscillator product in (2). The rest of this report gives that derivation:

1. encode K by seven affine walls;
2. rewrite the HFT quotient on the common D_4 torus;
3. telescope the Green–Schwarz product using a finite triality identity;
4. prove that the infinite high-mode tail is automatically safe;
5. check the fixed low-mode residue exactly.

This closes one major analytic premise of the trajectory–kernel programme. It does not yet classify all moving circuits or prove all-level integral packetization.

7.5 Programme at a glance

The route can be summarized without the finite computational detail as follows. A "complete" entry in the final column means complete for the mathematical statement indicated there, not completion of the Bars problem.

Table 2: Conceptual progression of the trajectory–kernel programme.

stage	new object or theorem	consequence
Bars	cumulative target R_n^9	formulate the minimal positive D_5-lift existence problem
HFT	covariant series $R(q)$	replace order-by-order oscillator construction by an exact all-orders B_4 generating function
TK1	Regge columns and reduced depth δ	raw multiplicity shifts are useful but not globally valid
TK2	boundary-corrected kernel shift	exact transport acts on restriction relations rather than on raw packets
TK3–5	positive fibres and chamber/core sources	establish finite existence and strong nonuniqueness; simple minimization is not canonical
TK6a	principal kernel ideal $(16_+ - 16_-)$	settle the global virtual kernel while isolating positivity as a semiring problem
TK6B.1	packet cone, semigroup and Hilbert basis	reduce finite integral liftability to facet inequalities in the level-11 closure
TK6B.2–4	Gale circuits and universal shells	replace thousands of facets by primitive weighted relations in a local collision hypergraph
TK6B.5b/c	Möbius pullback and boundary tails	turn facets into local HFT multiplicity inequalities and explain why circuit lists move with the cutoff
TK6B.5d/e	stable Pieri translation and hazard K	reduce the relevant transport inequalities to the single support implication $[V_\mu]F_d \neq 0 \Rightarrow d \geq K(\mu)$

stage	new object or theorem	consequence
TK6B.5f	Green–Schwarz telescope and finite low-mode core	prove that support implication at every degree

8 The seven-wall degree

8.1 B_4 labels and orthogonal coordinates

We write a dominant B_4 Dynkin label as

$$\mu = (\ell, m, r, s).$$

Its orthogonal coordinates are

$$x_1 = \ell + m + r + \frac{s}{2}, \qquad x_2 = m + r + \frac{s}{2},$$
$$x_3 = r + \frac{s}{2}, \qquad x_4 = \frac{s}{2},$$

with $x_1 \geq x_2 \geq x_3 \geq x_4 \geq 0$. Tensorial representations have integral x_i, while spinorial representations have half-integral x_i.

The original Pieri threshold $K(\mu)$ is piecewise affine. It is convenient to use a slightly stronger uniform function K_7, which agrees with K away from the wall $\ell = 0$ and may be stronger on that wall.

Definition 8.1 (Uniform seven-wall degree)**.** For $\mu = (\ell, m, r, s)$, define $K_7(\mu)$ as the maximum of

$$\begin{aligned}
W_1 &= 2\ell + 3m + 6r + 5s - 7,\\
W_2 &= 2\ell + 3m + 6r + 4s - 3,\\
W_3 &= 2\ell + 3m + 6r + 3s - 2,\\
W_4 &= 2\ell + 3m + 2s,\\
W_5 &= 2\ell + 5m + 6r + 3s - 4,\\
W_6 &= 2\ell + 5m + 7r + 4s - 5,\\
W_7 &= 2\ell + 5m + 9r + 5s - 9.
\end{aligned}$$

The first four walls are the exterior-chamber degree $E(\mu) - 1$; the last three are the extra unstable-Pieri hazards. Since

$$K_7(\mu) \geq K(\mu),$$

it is enough to prove the stronger implication

$$[V_\mu]F_d \neq 0 \quad \Longrightarrow \quad d \geq K_7(\mu).$$

For later use, Table 3 gives the same seven walls in orthogonal coordinates.

Table 3: The seven walls in orthogonal coordinates.

wall	coefficient vector	affine expression
W_1	$(2,1,3,4)$	$2x_1+x_2+3x_3+4x_4-7$
W_2	$(2,1,3,2)$	$2x_1+x_2+3x_3+2x_4-3$
W_3	$(2,1,3,0)$	$2x_1+x_2+3x_3-2$
W_4	$(2,1,-3,4)$	$2x_1+x_2-3x_3+4x_4$
W_5	$(2,3,1,0)$	$2x_1+3x_2+x_3-4$
W_6	$(2,3,2,1)$	$2x_1+3x_2+2x_3+x_4-5$
W_7	$(2,3,4,1)$	$2x_1+3x_2+4x_3+x_4-9$

9 Changing to the Green–Schwarz oscillator product

9.1 Why D_4 is the useful intermediate language

The covariant HFT series is naturally organized as a $B_4 = \mathrm{Spin}(9)$ character. The oscillator product is simpler after restriction to the common maximal torus of $D_4 = \mathrm{Spin}(8)$. Let

$$V = 8_v, \qquad S = 8_s, \qquad C = 8_c,$$

where S is the chirality chosen for the Green–Schwarz fermionic oscillators and C is the other half-spinor. Exchanging S and C is an outer automorphism of D_4; no physical chirality choice for a Spin(10) lift is being made at this stage.

Define the finite exterior polynomials

$$D(u) = \sum_{j=0}^{8} (-u)^j \Lambda^j V, \qquad E(u) = \sum_{j=0}^{8} u^j \Lambda^j S,$$

and set

$$H = V + S.$$

The inverse $D(u)^{-1}$ is the symmetric-power series of V.

9.2 The finite triality identity

Lemma 9.1 (One-mode divisibility)**.** *In the exact representation ring of D_4,*

$$E(u) - D(u) = uHB(u),$$

where

$$B(u) = 1 + (C-\mathbf{1})u^2 + (V-S)u^3 + (C-\mathbf{1})u^4 + u^6.$$

Proof. Both sides are polynomials of degree at most eight. Expand the exterior powers of 8_v and 8_s. Triality identifies the relevant vector and spinor exterior powers, while Hodge duality pairs degrees j and $8-j$. After subtracting, the constant terms cancel and every remaining coefficient is divisible by $H = V + S$. The quotients, in increasing degree, are

$$\mathbf{1}, \quad 0, \quad C-\mathbf{1}, \quad V-S, \quad C-\mathbf{1}, \quad 0, \quad \mathbf{1}.$$

This is exactly $B(u)$. The implementation verifies all nine coefficients as identities of Weyl characters. □

The identity is a finite representation-ring form of the generalized Jacobi identity relating RNS and Green–Schwarz oscillator descriptions. Its main advantage is that it performs the ground-multiplet division without expanding an infinite virtual Koszul sum.

9.3 Telescoping the infinite product

Put

$$A(u) = \frac{E(u)}{D(u)}.$$

The lemma gives

$$A(u) - 1 = \frac{uHB(u)}{D(u)}.$$

Let

$$P(q) = \prod_{m\geq 1} A(q^m)$$

be the Green–Schwarz oscillator product. For a finite product one has the elementary telescoping identity

$$\prod_{m=1}^{N} A_m - 1 = \sum_{n=1}^{N} (A_n - 1) \prod_{m=1}^{n-1} A_m.$$

Substituting $A_n = A(q^n)$ and using the one-mode identity gives, coefficient by coefficient in q,

$$\frac{P(q) - 1}{qH} = \sum_{n\geq 1} q^{n-1} \frac{B(q^n)}{D(q^n)} \prod_{m=1}^{n-1} A(q^m).$$

There is no convergence issue: the coefficient of q^d sees only finitely many modes.

The B_4 vector restricts to $\mathbf{1} + V$ on D_4. Removing its level-one symmetric tower therefore multiplies by

$$\mathrm{PE}(-q(\mathbf{1} + V)) = (1 - q)D(q).$$

We obtain the central exact identity.

Proposition 9.2 (Green–Schwarz telescope for the remainder)**.** *On the common D_4/B_4 torus,*

$$F(q) = (1 - q) \sum_{n\geq 1} q^{n-1} B(q^n) \frac{\prod_{m=1}^{n-1} E(q^m)}{\prod_{m=2}^{n} D(q^m)}. \tag{20}$$

This is the decisive change of viewpoint. The universal multiplet has been divided out exactly, with the required cancellations already incorporated. Every summand has a distinguished first nontrivial mode n, which we call its *pivot*.

10 Returning from D_4 to B_4

10.1 The finite Weyl-numerator map

Let A_ρ denote a Weyl alternant. Since the roots of B_4 not present in D_4 are the short roots $\pm e_i$, the Weyl denominators obey

$$\frac{A_{\rho_{B_4}}}{A_{\rho_{D_4}}} = \prod_{i=1}^{4} \left(e^{e_i/2} - e^{-e_i/2} \right).$$

For a D_4 irrep with highest weight λ,

$$A_{\rho_{B_4}}\chi_\lambda^{D_4} = \frac{A_{\rho_{B_4}}}{A_{\rho_{D_4}}}A_{\lambda+\rho_{D_4}}.$$

The right side has at most $192 \cdot 16$ signed monomials before equal terms are collected: 192 elements of the D_4 Weyl group and 16 sign choices from the short-root product. Thus the conversion is finite and independent of the size of λ.

10.2 Chirality folding does not increase the wall degree

Write a dominant D_4 highest weight in orthogonal coordinates as

$$\lambda = (a, b, c, d), \qquad a \geq b \geq c \geq |d|.$$

Its chirality fold is the dominant B_4 weight

$$\widehat{\lambda} = (a, b, c, |d|).$$

Lemma 10.1 (Componentwise containment)**.** *If a dominant B_4 term $\nu = (y_1, y_2, y_3, y_4)$ occurs in $A_{\rho_{B_4}}\chi_\lambda^{D_4}$, then*

$$y_i \leq \widehat{\lambda}_i \qquad (1 \leq i \leq 4).$$

Proof. The doubled Weyl-numerator terms have the form

$$w(2\lambda + 2\rho_{D_4}) + \varepsilon, \qquad \varepsilon_i \in \{\pm 1\}.$$

Take absolute values and order the four coordinates decreasingly. There is only one coordinate originating from each entry of $\widehat{\lambda} + \rho_{D_4}$. Consequently the ith order statistic is no larger than

$$\widehat{\lambda}_i + (\rho_{D_4})_i + \frac{1}{2} = \widehat{\lambda}_i + (\rho_{B_4})_i.$$

Subtracting ρ_{B_4} gives the claim. □

Six of the seven walls in Table 3 have nonnegative orthogonal-coordinate coefficients, so componentwise containment immediately controls them. Only W_4 needs attention.

Lemma 10.2 (The exceptional wall)**.** *Every dominant term ν in the Weyl numerator satisfies*

$$K_7(\nu) \leq K_7(\widehat{\lambda}).$$

Proof. For the exceptional wall, dominance of ν gives

$$\begin{aligned} W_4(\nu) &= 2y_1 + y_2 - 3y_3 + 4y_4 \\ &\leq 2a + b + |d|, \end{aligned}$$

because $-3y_3 + 4y_4 \leq y_4 \leq |d|$. If $c = |d|$, the last expression is exactly $W_4(\widehat{\lambda})$. If $c > |d|$, the two coordinates lie in the same weight lattice, so $c \geq |d| + 1$. Hence

$$2a + b + |d| \leq 2a + b + 3c - 2 = W_3(\widehat{\lambda}).$$

The maximum of all seven source walls therefore bounds every target wall. □

This lemma explains precisely what chirality does and does not do here. Different D_4 chiral representatives may contribute differently, but after folding, none can create a B_4 term of larger filtered degree. No choice of a $D_5 = \mathrm{Spin}(10)$ parent has entered the argument.

11 Why only finitely many oscillator modes remain

11.1 Vector and spinor Lipschitz constants

The seven walls provide uniform tensor-product estimates.

Lemma 11.1 (One-factor degree increase)**.** *Let α be a dominant D_4 weight and fold all D_4 chiralities before evaluating K_7.*

1. *Every constituent of $V_\alpha \otimes V$ has degree at most $K_7(\widehat{\alpha}) + 4$.*
2. *Every constituent of $V_\alpha \otimes S$ has degree at most $K_7(\widehat{\alpha}) + 5$.*

Proof. A vector weight changes one orthogonal coordinate by ± 1 before Weyl dominantization. The largest absolute coefficient in Table 3 is 4. A spinor weight changes all four coordinates by $\pm\frac{1}{2}$. Half the largest sum of absolute wall coefficients is

$$\frac{1}{2}(2 + 1 + 3 + 4) = 5.$$

Taking the maximum over walls and Weyl images preserves these bounds. □

Repeated tensoring gives an increase of at most $4r$ for a constituent of $V^{\otimes r}$ and at most $5j$ for a constituent of $S^{\otimes j}$. Since $\mathrm{Sym}^r V \subset V^{\otimes r}$ and $\Lambda^j S \subset S^{\otimes j}$, the same upper bounds apply to the oscillator factors.

11.2 The safe tail

In a bosonic mode m, the degree charge of $\mathrm{Sym}^r V$ is mr. Thus every vector mode $m \geq 4$ pays at least the maximum possible increase $4r$. Similarly, every fermion mode $m \geq 5$ pays at least $5j$ for $\Lambda^j S$.

The polynomial $B(q^n)$ is also harmless for $n \geq 3$:

power in B	possible non-scalar factor	degree charge for $n \geq 3$
2	C	at least 6
3	V or S	at least 9
4	C	at least 12

The possible filtered increases are at most 5, 5 and 5 respectively. Negative coefficients such as $-\mathbf{1}$ or $-S$ do not change a support bound; they change only signed multiplicities.

We have therefore reduced the entire infinite product to

- fermionic modes 1, 2, 3 and 4;
- bosonic vector modes 2 and 3;
- the exceptional pivot polynomials $B(q)$ and $B(q^2)$.

This reduction is uniform in the final string level.

12 The two low vector modes

12.1 Harmonic decomposition and the hazard transform

Let H_k be the D_4 harmonic one-row representation. Recall

$$\mathrm{Sym}^r V = H_r \oplus H_{r-2} \oplus H_{r-4} \oplus \cdots .$$

For an upper support estimate, a trace term is easier than its harmonic label suggests: it pays the oscillator degree of r but has harmonic degree $k \le r$. It is therefore enough to charge mk to H_k in mode m.

For a folded source α, define the exact mode-two hazard

$$H_2(\alpha) = \max_{\substack{k>0 \\ V_\beta \subset \overline{V}_\alpha \otimes H_k}} \left(K_7(\widehat{\beta}) - 2k \right).$$

This is finite: after the first-row stable range begins, increasing k only translates the leading coordinate and does not enlarge the charged degree.

12.2 Pieri-polytope optimization

The orthogonal horizontal-strip Pieri rule can be expressed using an intermediate partition ξ. Both the source-to-ξ skew diagram and the target-to-ξ skew diagram are horizontal strips. The number of boxes removed and added bounds the harmonic degree k. In the spinorial coset, subtracting the common half-spin shift gives the same interlacing inequalities.

Each wall is affine in the target rows, while the degree penalty is affine in the source, intermediate and target rows. Maximizing over the Pieri polytope is therefore an elementary endpoint calculation: every variable is put at one of its two interlacing endpoints according to the sign of its coefficient. This gives the certified upper closures in Table 4.

Table 4: Affine upper closure under vector mode 2. A row $(p_1, p_2, p_3, p_4) - c$ denotes $p_1\ell + p_2 m + p_3 r + p_4 s - c$.

input wall	mode-two upper closure
$(2,3,6,5) - 7$	$(2,3,8,5) - 7$
$(2,3,6,4) - 3$	$(2,4,6,4) - 3$
$(2,3,6,3) - 2$	$(2,4,6,3) - 2$
$(2,3,0,2)$	$(2,3,0,2)$
$(2,5,6,3) - 4$	$(3,5,6,3) - 4$
$(2,5,7,4) - 5$	$(3,5,7,4) - 5$
$(2,5,9,5) - 9$	$(3,7,9,5) - 9$

Let $\overline{H}_2$ be the maximum of the seven right-hand entries. Then

$$H_2(\alpha) \le \overline{H}_2(\alpha).$$

Applying the same endpoint optimization to mode 3 leaves those seven upper pieces and adds one new piece,

$$4\ell + 7m + 9r + 5s - 9.$$

Thus the joint exact hazard H_{23} obeys

$$H_{23}(\alpha) \le \overline{H}_{23}(\alpha) := \max\left\{\overline{H}_2(\alpha), 4\ell + 7m + 9r + 5s - 9\right\}. \tag{21}$$

Only the upper bound is needed for the proof. Direct D_4 character products attain it throughout the regression simplex, including examples in which mode 3 genuinely raises the mode-two hazard. These comparisons protect against convention or chirality errors, but they are not being extrapolated to establish the theorem.

13 The finite low-mode certificate

13.1 Five pivot classes

Denote the nth summand of (20) by

$$T_n(q) = (1-q)q^{n-1}B(q^n)\frac{\prod_{m=1}^{n-1} E(q^m)}{\prod_{m=2}^{n} D(q^m)}.$$

After applying the safe-tail estimates, only five cases remain.

Table 5: Finite data retained in each pivot class.

pivot	base degree	fermionic core	remaining vector/core factor
1	0	none	$(1-q)B(q)$
2	1	$E(q)$	$B(q^2)/D(q^2)$
3	2	modes 1–2	vector modes 2–3
4	3	modes 1–3	vector modes 2–3
$n \ge 5$	at least 4	modes 1–4	vector modes 2–3

For $n \ge 5$, replacing the actual base degree $n-1$ by 4 is the worst-case choice. All omitted fermion modes are at least 5, all omitted vector modes are at least 4, and $B(q^n)$ is safe.

13.2 Why first occurrence is sufficient

Expand each finite fermion core exactly in irreducible D_4 characters. If the same irrep occurs at several degrees, retain only its first occurrence. The representation has the same low-vector hazard at every occurrence, while later copies have more degree available. This turns the three nontrivial fermion cores into only

$$20, \qquad 50, \qquad 105$$

distinct first-occurrence records for pivots 3, 4 and $n \ge 5$.

For each source α of first degree f, the required calculation is simply

$$\text{margin} = (\text{base degree}) + f - \overline{H}_{23}(\alpha).$$

Pivots 1 and 2 retain the signed coefficients of B explicitly. For pivot 2 the mode-two upper hazard $\overline{H}_2$ is used. The factor $(1-q)$ creates a second copy one degree later, so it can never worsen support.

13.3 Certificate result

Table 6: Complete low-mode certificate.

pivot class	records	minimum margin	failures
1	8	0	0
2	140	0	0
3	20	0	0
4	50	0	0
$n \geq 5$	105	1	0
total	323	0	0

All decompositions are exact Weyl-character calculations over the integers. The virtual signs in B are retained. No protected Spin(10) lift, no level-1–10 answer and no numerical optimization solver enters the certificate.

14 Proof of the all-orders support theorem

We can now assemble the argument.

Theorem 14.1 (All-orders HFT remainder support)**.** *For every $d \geq 0$ and every dominant B_4 label μ,*

$$[V_\mu]F_d \neq 0 \quad \Longrightarrow \quad d \geq K_7(\mu) \geq K(\mu).$$

Proof. Use the exact telescoping expansion (20) and fix one pivot n.

First decompose the finite fermion core retained in Table 5. For each resulting D_4 source, the certificate in Table 6, together with (21), proves degree at least K_7 after applying vector modes 2 and 3. Pivots 1 and 2 are included directly in the same certificate.

Every remaining vector mode has index at least 4, so its oscillator degree pays for the maximum filtered increase. Every remaining fermion mode has index at least 5 and is safe for the same reason. The factor $B(q^n)$ is safe for $n \geq 3$, while its exceptional cases $n = 1, 2$ were included in the finite certificate. Multiplication by $(1 - q)$ either preserves the degree or raises it by one.

Thus every D_4 irrep occurring in every coefficient of every pivot summand has degree at least the folded value of K_7. The Weyl-numerator lemma shows that passage back to dominant B_4 characters cannot increase K_7. Hence each individual telescoping summand has the desired support, and so does their signed sum $F(q)$.

Finally, $K_7 \geq K$, proving the original Pieri-hazard support theorem. □

Corollary 14.2 (Stable-range premise)**.** *Every contribution of the virtual HFT remainder lies in the stable range required by the consecutive harmonic and symmetric-power Pieri translation theorem. The new top-degree contribution also lies outside the translated exterior-chamber target.*

The corollary is valid contribution by contribution, including spinorial sources and negative coefficients in the virtual remainder.

15 Why the earlier Koszul route failed

There is a formally correct B_4 identity

$$F_d = \sum_{j=0}^{9} (-1)^j \Lambda^j(V_9) \otimes R_{d-j}.$$

It is tempting to bound every tensor product on the right separately using minimum physical levels. That strategy is too strong because the sum is virtual. An unsafe irrep can occur in one Koszul term and be cancelled by an equal irrep in another term.

The simplest symptom is already visible near the vacuum: a term $-V_9 q$ violates the desired bound term by term, but is cancelled by the corresponding positive contribution. In the standard finite regression, the generic Koszul-transition test produces 165 such obstructions. These are not counterexamples to the theorem; they are counterexamples to a proof method which discarded essential cancellations.

The Green–Schwarz telescope succeeds because the cancellation required for division by the universal multiplet is performed algebraically first. Only then is a support estimate applied. This distinction is the conceptual reason the new proof closes while the principal-embedding shortcut does not.

16 Independent checks

The proof is level-independent, but several exact finite comparisons guard against implementation errors.

Table 7: Standard regression checks. These checks support the implementation; they are not substitutes for the all-orders argument.

check	cases or terms	failures
one-mode triality coefficients	9	0
telescoping reconstruction, degrees 0–5	6	0
separate telescoping support terms	33	0
finite Weyl/Dirac fold regression	30	0
direct low-vector hazard comparisons	15	0
finite low-mode certificate	323	0
observed HFT remainder irreps through physical level 13	176	0

The telescope reconstructs the independently derived B_4 remainder weight-for-weight through degree 5. A larger direct test of the total D_4 irrep support has also passed through degree 24. These observations were useful in finding the proof, but the finite-mode reduction is what removes dependence on a level cutoff.

17 What has and has not been proved

17.1 What is now closed

The theorem supplies an analytic, all-degree answer to the question which motivated this report:

- higher oscillator levels cannot introduce a new support violation;
- chirality folding at the D_4 stage cannot increase the required B_4 degree;
- every virtual remainder contribution lies in the stable Pieri range;
- the result is baseline-free and does not depend on a selected Spin(10) representative.

17.2 What remains for Bars III.B

The Bars claim concerns positive integral Spin(10) lifts of the cumulative targets R_n^9. The support theorem controls one systematic family of local inequalities, but the complete finite-horizon lift cone has moving weighted circuits. The remaining work is to do at least one of the following:

1. prove that the stable Pieri transport assembles all moving boundary-tail circuits into a recursively controlled family;
2. construct an all-level nonnegative integral packetization directly;
3. prove that another graded construction, such as the BBMS candidate, telescopes to a positive cumulative lift at every level.

Even if all rational circuit inequalities are controlled, integrality or saturation remains a logically separate issue.

17.3 What the proof says about chirality

The argument deliberately folds the two D_4 chiralities only for the purpose of a support estimate. It shows that the support theorem is insensitive to that choice. It does *not* prove that different Spin(10) chiral lifts are physically equivalent, nor does it select among the distinct positive completions seen from level 6 onward. That question belongs to the later global lift problem.

18 Reproduction and implementation boundary

The proof implementation is

`trajectory_kernel/hft_analytic_support.py`.

From the project root, the complete standard run is

```
./bin/sage-run -m trajectory_kernel.hft_analytic_support \
  --max-level 13 \
```

```
  --maximum-label-sum 2 \
  --telescoping-degree 5 \
  --core-pivot 3 --core-pivot 4 --core-pivot 5 --progress
```

It reports, among other checks,

```
Finite low-mode pivots/sources/failures: (3, 4, 5)/175/0
Finite core minimum margin: 0
Complete finite certificate: True
Protected SO(10) baseline imported: NO
```

The parameter `-max-level` affects only comparison with explicitly expanded HFT coefficients. It does not change the finite proof domain. The proof modules do not import any of the protected level-1–10 Spin(10) baseline files.

A A compact logical dependency map

For reference, the proof can be read as the following chain.

1. The seven-wall degree K_7 implies the exact Pieri threshold K.
2. Triality gives $E - D = uHB$.
3. That identity telescopes the Green–Schwarz oscillator product and gives equation (20).
4. The D_4 Weyl numerator maps to B_4 without increasing K_7.
5. Vector and spinor Lipschitz constants close all modes except fermions 1–4 and vectors 2–3.
6. Pieri-polytope optimization bounds vector modes 2–3 by eight affine walls.
7. The remaining fermionic first occurrences form a 323-record exact certificate.
8. Therefore every coefficient of every pivot summand, and hence every coefficient of $F(q)$, satisfies degree at least K_7.

B Notation glossary

symbol	meaning
B_4	Spin(9), the massive little group in ten dimensions
D_4	Spin(8), used for the transverse Green–Schwarz oscillator product
D_5	Spin(10), the desired parent group for Bars lifts
r_n^9	chiral massive-primary B_4 character at physical level n
R_n^9	cumulative Bars target $\sum_{i=1}^{n} r_i^9$
$R(q)$	HFT massive-primary series after universal massive-multiplet division
$F(q)$	virtual remainder after removing the level-one B_4 vector tower
P_λ	branching packet $\mathrm{Res}_{B_4}^{D_5} V_\lambda$
U	unitriangular matrix of all-chiral reference packets

$c = U^{-1}m$	integral, generally signed, triangular section coordinates
J	global restriction-kernel generator $16_+ - 16_-$
facet	codimension-one face, or primitive defining inequality, of the packet cone
Hilbert basis	minimal integral generators of all lattice points in a rational cone
Gale vector	row of an integral kernel-basis matrix for the packet configuration
positive circuit	minimally supported nonnegative relation among Gale vectors
shell	six-term local kernel identity from consecutive interlacing slices
Regge shift	simultaneous $n \mapsto n+1$ and $\ell \mapsto \ell+1$
$\rho = n - \ell$	elementary invariant of the Regge shift
$\delta = n - \ell - 3m$	support-adapted reduced depth; subtracting $3m$ absorbs the level cost of the second Dynkin label
upper ℓ-chamber	high-ℓ supported region where multiplicity is inherited from a lower-level Regge predecessor
transport boundary	lattice wall separating the upper ℓ-chamber from the sector where that transport rule is unavailable
branching boundary	interlacing face $\mu_1 = \lambda_2$ left behind when a branching packet is shifted
residual core	supported labels reached by none of the admissible ℓ-, m- or r-transport identities
stable Pieri range	region $\lambda_1 \geq \mu_1$ where one-row tensor multiplicities translate exactly
V, S, C	D_4 vector, chosen oscillator spinor and opposite spinor
$D(u)$	alternating exterior polynomial of V
$E(u)$	ordinary exterior polynomial of S
$B(u)$	finite triality quotient in $E - D = u(V+S)B$
$K(\mu)$	exact unstable-Pieri source threshold from TK6B.5e
$K_7(\mu)$	uniform seven-wall strengthening of K used in this proof
H_k	harmonic one-row representation of degree k
pivot n	first oscillator mode selected by the telescoping identity

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